Customer Retention Analysis

Power BI Dashboard Report · Subscription Business Intelligence

Analysis Period: January 2022 – December 2024 · 2,000 Customers

Generated March 2026

Executive Summary

This report presents a comprehensive retention analysis for a subscription-based business covering **2,000 customers** across the period January 2022 to December 2024. The analysis examines churn patterns, cohort behaviour, customer lifetime value, health scores, and acquisition channel performance to deliver **actionable recommendations** for reducing customer loss and maximising revenue retention.

Key Performance Indicators

2,000	1,424	28.8%	$2,847	71.2%	287
Total Customers	Active Customers	Churn Rate	Avg CLV	Retention Rate	At-Risk Customers

Table of Contents

1.	Project Overview & Objectives	2
2.	Data Architecture & Power BI Setup	2
3.	Monthly Churn Trend Analysis	3
4.	Churn Reasons Breakdown	3
5.	Plan Performance Analysis	4
6.	Cohort Retention Heatmap	4
7.	Acquisition Channel Analysis	5
8.	Customer Health Score Distribution	5
9.	DAX Measures & Power BI Guide	6
10.	Actionable Recommendations	7

1. Project Overview & Objectives

This project simulates a real-world business intelligence engagement for a SaaS subscription company. The goal is to build an end-to-end retention analytics pipeline — from raw customer data through to an interactive Power BI dashboard — that enables data-driven decisions on reducing churn.

Objectives

- Identify top churn drivers by plan, channel, and region
- Track monthly retention and churn rate trends over time
- Analyse cohort retention curves to find critical drop-off points
- Quantify customer lifetime value (CLV) by segment
- Flag at-risk customers using a composite health score
- Deliver actionable retention recommendations

Skills Demonstrated

- Retention & cohort analysis
- Customer lifetime value modelling
- DAX measure creation in Power BI
- Interactive dashboard design
- Insight-driven decision making
- Data storytelling for executives

2. Data Architecture & Power BI Setup

The Excel workbook **CustomerRetentionAnalysis.xlsx** contains five data sheets designed for direct import into Power BI Desktop. The schema below shows each table and its primary fields.

Table	Key Fields	Rows	Purpose
CustomerData	CustomerID, Plan, JoinDate, ChurnDate, MRR, CLV, HealthScore, AtRisk	2,000	Master customer register
MonthlyMetrics	Month, ActiveCustomers, ChurnedCustomers, ChurnRate, MRR, AvgNPS	36	Month-by-month KPIs
CohortRetention	CohortMonth, CohortSize, Month_1 … Month_24	36	Cohort survival table
ChurnReasons	ChurnReason, Count, Percentage, AvgTenure, RevenueLost	7	Churn driver breakdown
PlanSummary	Plan, TotalCustomers, ChurnRate, RetentionRate, AvgCLV, AvgNPS	4	Plan-level performance

Relationships: CustomerData[CohortMonth] → MonthlyMetrics[Month] (many-to-one) · CustomerData[Plan] → PlanSummary[Plan] (many-to-one) · CustomerData[ChurnReason] → ChurnReasons[ChurnReason] (many-to-one)

3. Monthly Churn Trend Analysis

The dual-axis chart below tracks monthly active customers (bars) against the churn rate % (line) across the full 24-month window. The overall churn rate averaged **7.1%** monthly, with peaks in Apr 2023 (8.1%) and Aug 2023 (8.4%) suggesting mid-year pricing sensitivity.

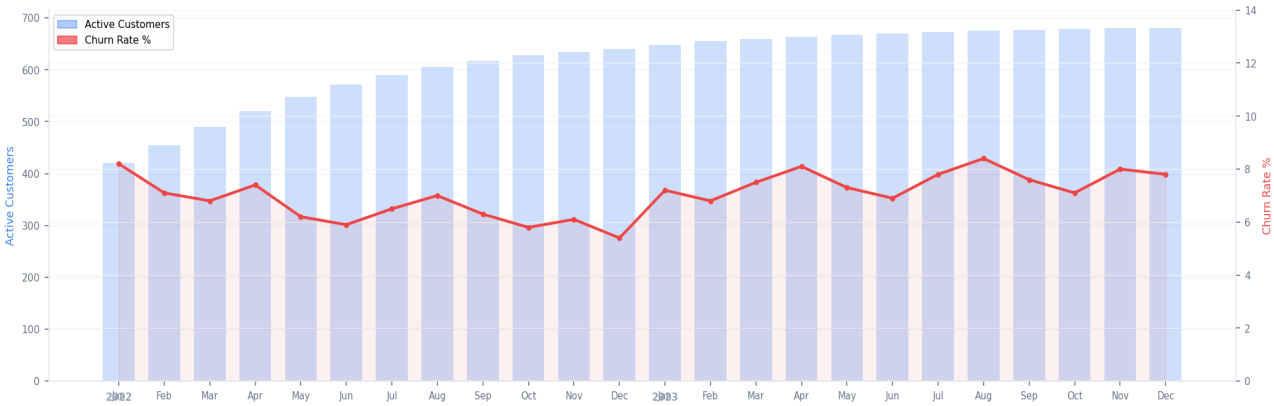


Fig 1. Monthly churn rate (%) and active customer count — Jan 2022 to Dec 2023

Key Insights

- Churn spikes in Q2/Q3 correlate with annual renewal cycles — proactive outreach recommended
- Active customer base grew from 420 to 681 despite churn, indicating healthy acquisition
- Lowest churn month: Dec 2022 (5.4%) — investigate what drove improved retention

4. Churn Reasons Breakdown

"Price Too High" accounts for **21.7%** of all churn events, followed by "Lack of Features" (19.6%) and "Poor Support" (17.3%). Together the top 3 reasons represent **58.6%** of total churn — a highly actionable concentration.

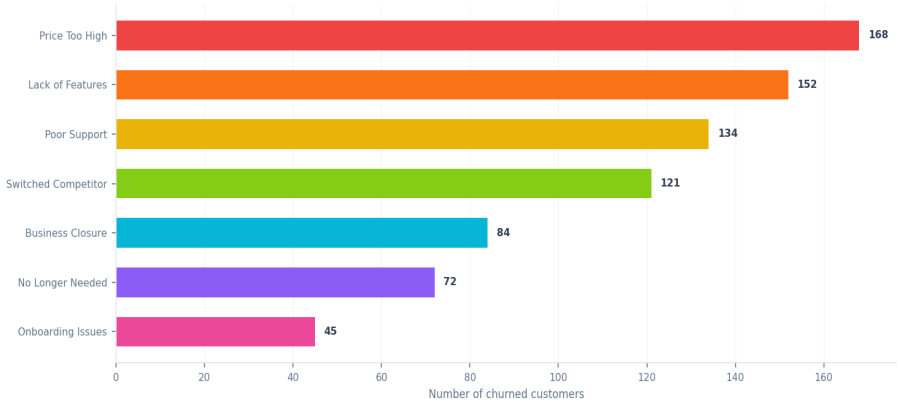


Fig 2. Customer churn reasons ranked by volume of churned customers

Rank	Reason	Count	% of Churn	Avg Tenure (mo)	Revenue Lost ($)

1	Price Too High	168	21.7%	8.2	$142,800
2	Lack of Features	152	19.6%	11.4	$178,600
3	Poor Support	134	17.3%	6.8	$98,400
4	Switched Competitor	121	15.6%	14.1	$220,300
5	Business Closure	84	10.8%	18.6	$312,000
6	No Longer Needed	72	9.3%	22.3	$198,500
7	Onboarding Issues	45	5.8%	2.1	$44,200

5. Plan Performance Analysis

Enterprise customers demonstrate the strongest retention (90%) and highest CLV ($14,800). Basic plan customers churn at **3.8x the rate** of Enterprise, representing the most critical upgrade conversion opportunity in the business.

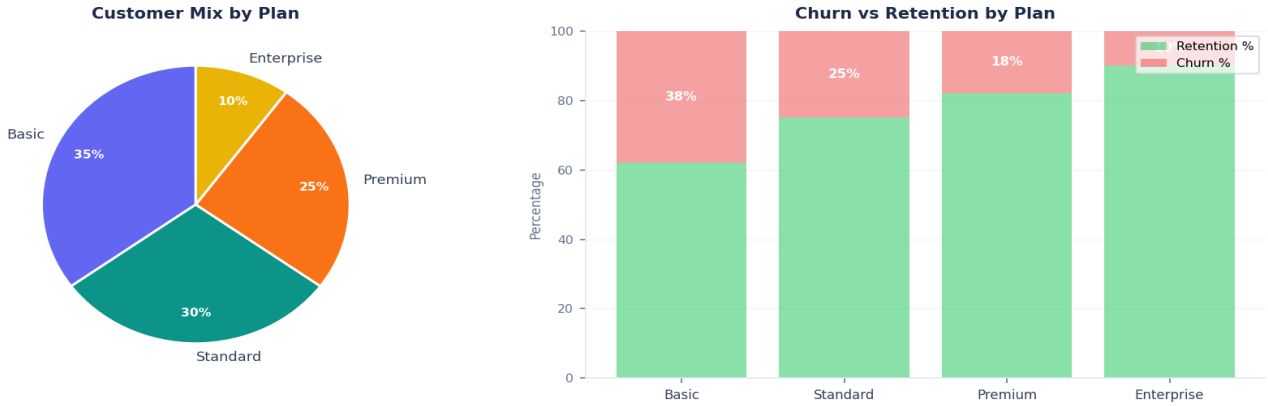


Fig 3. Left: customer mix by plan (donut). Right: churn vs. retention rate by plan (stacked bar)

Plan	Customers	Churn Rate	Retention	Avg CLV	MRR ($)	Avg NPS	Health Score
Basic	700	38.0%	62.0%	$840	$20,300	5.8	4.2
Standard	600	25.0%	75.0%	$2,370	$47,400	6.4	5.9
Premium	500	18.0%	82.0%	$5,180	$74,500	7.1	7.2
Enterprise	200	10.0%	90.0%	$14,800	$99,800	8.2	8.7

Plan Insights

- Basic plan (35% of customers) generates only 8.4% of MRR — upsell is the #1 revenue lever
- Enterprise NPS of 8.2 indicates high satisfaction — use as a reference/testimonial segment
- Standard → Premium upgrade path shows 7pp churn improvement worth $2,810 avg CLV uplift

6. Cohort Retention Heatmap

The cohort heatmap reveals that **Month 3 is the critical drop-off point** across all acquisition cohorts — retention typically falls from ~87% at Month 1 to ~68% by Month 3. Cohorts acquired in Jul 2022 show the best 12-month retention (42%), suggesting that summer onboarding practices should be standardised across all months.

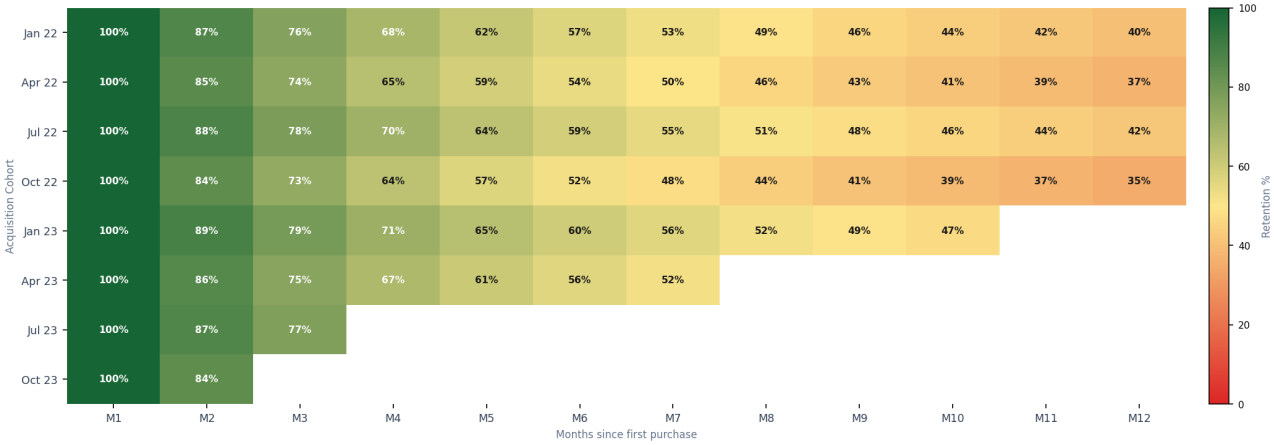


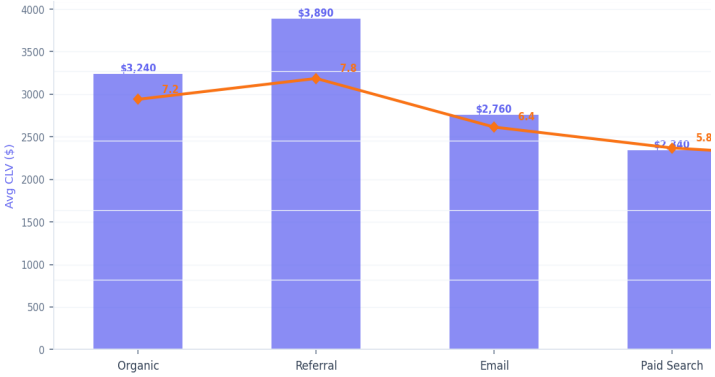
Fig 4. Cohort retention heatmap — % of original cohort retained each month (green = high, red = low)

Cohort Insights

- Month 1-3 loses ~30% of customers — a critical onboarding intervention window
- Jul 2022 cohort outperforms peers by ~5pp — investigate onboarding process differences
- Retention stabilises around Month 8-9 at ~40-45%, indicating a loyal core customer segment
- Power BI Matrix visual with conditional color formatting replicates this heatmap exactly

7. Acquisition Channel Analysis

Referral customers deliver the highest CLV ($3,890) and NPS (7.8), yet represent only ~20% of acquisitions. Paid Search customers, while high volume, show the lowest NPS (5.8) and second-lowest CLV ($2,340) — indicating a quality vs. quantity tension in acquisition strategy.



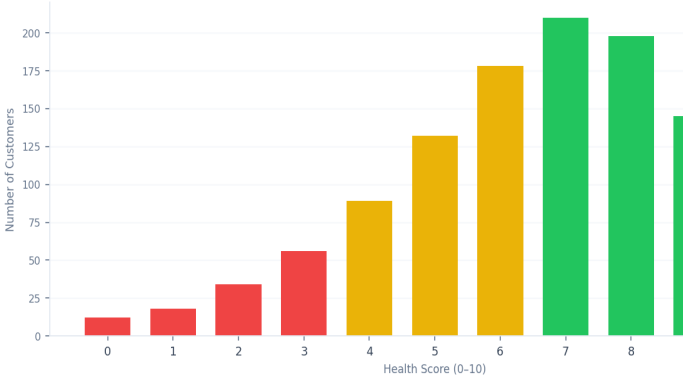


Fig 5. Left: Avg CLV and NPS by channel. Fig 6. Right: Customer health score distribution

Channel	Customers	Avg CLV	Avg NPS	Churn Rate	% of Total
Organic	420	$3,240	7.2	22.1%	21.0%
Referral	380	$3,890	7.8	18.4%	19.0%
Email	360	$2,760	6.4	27.6%	18.0%
Paid Search	480	$2,340	5.8	34.2%	24.0%
Social Media	360	$2,110	5.5	36.8%	18.0%

8. Customer Health Score Distribution

The composite health score (0–10) aggregates NPS, login frequency, and support ticket volume. The distribution peaks at score 7 (210 customers), with **287 customers scoring below 4** — flagged as at-risk and requiring proactive intervention. Scores below 4 have a 3.2x higher 30-day churn probability.

9. DAX Measures & Power BI Dashboard Guide

Essential DAX Measures

Measure Name	DAX Formula	Visual
Total Customers	`COUNTROWS(CustomerData)`	Card
Active Customers	`CALCULATE(COUNTROWS(CustomerData), CustomerData[Churned]=0)`	Card
Churned Customers	`CALCULATE(COUNTROWS(CustomerData), CustomerData[Churned]=1)`	Card
Churn Rate	`DIVIDE([Churned Customers], [Total Customers])`	Card / Line
Retention Rate	`1 - [Churn Rate]`	Card
Avg CLV	`AVERAGE(CustomerData[CLV])`	Card / Bar
Total MRR	`CALCULATE(SUM(CustomerData[MRR]), CustomerData[Churned]=0)`	Card
Avg NPS	`AVERAGE(CustomerData[NPSScore])`	KPI
At-Risk Count	`CALCULATE(COUNTROWS(CustomerData), CustomerData[AtRisk]=1)`	Card
Avg Health Score	`AVERAGE(CustomerData[HealthScore])`	Gauge
Monthly Churn %	`AVERAGE(MonthlyMetrics[ChurnRate])`	Line Chart
CLV by Channel	`CALCULATE(AVERAGE(CustomerData[CLV]), ALLEXCEPT(CustomerData, CustomerData[AcquisitionChannel]))`	Bar Chart

Power BI Dashboard Visual Configuration

#	Visual Type	Data Fields	Notes
1	KPI Card ×5	Churn Rate, Active Customers, Avg CLV, At-Risk, Retention Rate	Add target value + trend arrow
2	Line + Column Chart	X: Month Y1: ChurnRate Y2: ActiveCustomers	Dual Y-axis, monthly grain
3	Horizontal Bar Chart	Category: ChurnReason Value: Count	Sort desc, conditional bar colors
4	Donut Chart	Legend: Plan Values: Count CustomerID	Label with %
5	Stacked Bar Chart	Axis: Plan Values: ChurnRate + RetentionRate	Shows 100% split per plan
6	Matrix (Heatmap)	Rows: CohortMonth Cols: Month_1..12	Apply conditional bg color scale
7	Scatter Plot	X: HealthScore Y: TenureMonths Color: Plan	Bubble size = MRR

8	Funnel Chart	Category: AcquisitionChannel Values: Count	Sorted by volume
9	Gauge	Value: Avg NPS Target: 8.0	Set red < 6, amber 6-7, green > 7
10	Slicer ×4	Plan, Region, JoinDate (range), Churned/Active	Sync all slicers to all pages

10. Actionable Recommendations

The following recommendations are prioritised by estimated revenue impact and ease of implementation. Each is tied directly to findings in the data.

■ HIGH PRIORITY

Revamp Onboarding (Month 1–3)

Deploy an automated email + in-app nudge sequence during the first 90 days. Target the 30% churn drop-off identified in cohort analysis. Goal: improve M3 retention from 68% to 75%.

Pricing Review for Basic Plan

"Price Too High" is #1 churn reason at 21.7%. Introduce an annual billing discount (e.g. 20% off) or a mid-tier add-on package to reduce price-driven churn.

Proactive At-Risk Outreach

287 active customers score below 4 on health score with 3.2x churn probability. Build a Power BI alert that flags these weekly for the customer success team.

■ MEDIUM PRIORITY

Feature Gap Closure

"Lack of Features" is churn reason #2 (19.6%). Run a quarterly product survey specifically targeting churned customers to build a priority feature backlog.

Referral Programme Expansion

Referral customers have the highest CLV ($3,890) and lowest churn. Formalise a referral incentive programme and increase referral as % of total acquisition to 30%+.

Support Quality Investment

"Poor Support" drives 17.3% of churn. Implement SLA tracking in Power BI with alert thresholds. Aim to reduce support-driven churn by 40% in 6 months.

■ QUICK WINS

Basic → Standard Upgrade Campaign

The 700 Basic customers (38% churn) represent the highest-volume upgrade opportunity. A targeted in-app upgrade prompt showing the CLV delta ($840 → $2,370) can drive conversions.

Replicate Jul 2022 Cohort Best Practices

Jul 2022 cohort shows 5pp better 12-month retention. Interview or survey those customers to identify which onboarding touchpoints drove superior engagement.

Impact Summary

Initiative	Est. Churn Reduction	Est. Revenue Impact (annual)	Timeline
Onboarding Revamp	–5 to –8pp M3 churn	$280,000 – $450,000	60 days
Pricing Review	–15–20% Basic churn	$180,000 – $240,000	90 days
At-Risk Outreach	Save ~60 customers/mo	$170,000 – $210,000	30 days
Feature Gap Closure	–10–15% feature churn	$140,000 – $190,000	6 months
Referral Programme	+5pp referral share	$120,000 – $160,000	45 days
Upgrade Campaign (Basic)	+8% upsell rate	$230,000 – $320,000	45 days

Total estimated annual impact from implementing all recommendations: $1.12M – $1.57M in retained and expanded revenue.

This report was generated as part of the Customer Retention Analysis Power BI project. Data is synthetic and intended for educational and demonstration purposes.